

ସ୍ଥିର ଚାର୍ଜ

$$9 \times 10^9$$

$$F \propto q_1 q_2$$

$$F \propto \frac{1}{r^2}$$

$$F = K \frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$$F = \frac{1}{4\pi K \epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

$$(E_r) \rightarrow K = \frac{\epsilon}{\epsilon_0}$$

$$\epsilon_r = \frac{\epsilon_a}{\epsilon_0}$$

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

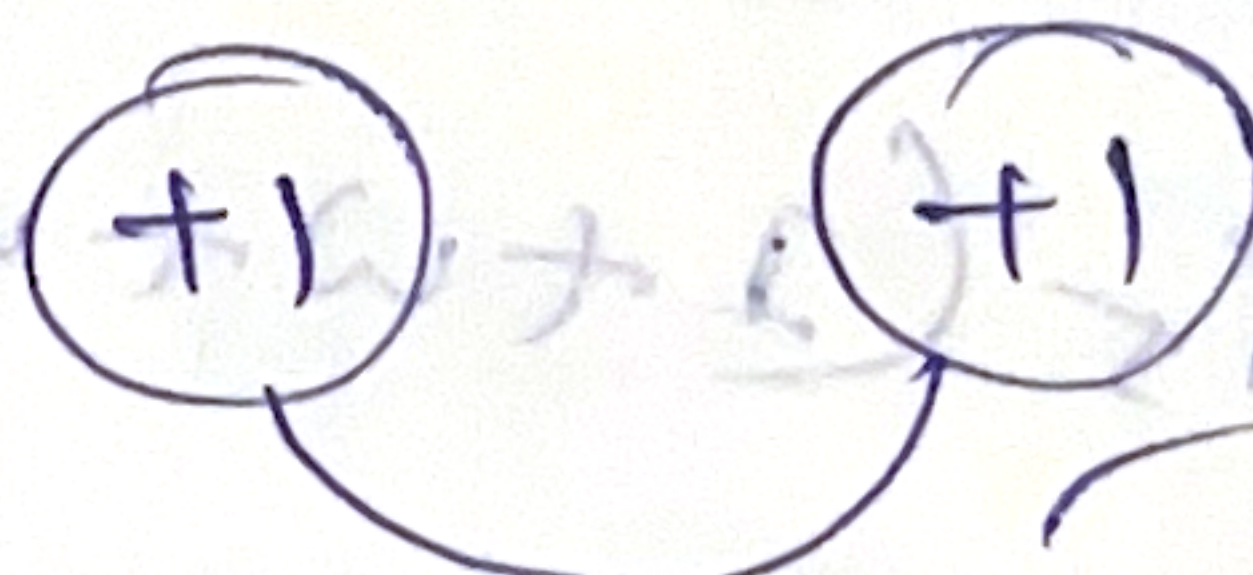
$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} \cdot \frac{r}{|r|}$$



$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^3} = \frac{1}{r^2}$$

$$F = K \cdot \frac{q_1 q_2}{r^2}$$

$$F = 9 \times 10^9 \text{ N}$$



+ $\rightarrow$ repulsive

- $\rightarrow$ attractive

(+)

$q = ne^-$

n

1.6×10^{-19}

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$2.4 \times 10^{-19} \text{ C}$

1.5×10^{-19}

(+)

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

$= 9 \times 10^9 \times$

उत्तर

$$E = - \frac{dV}{dx}$$

$\frac{V}{m}$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

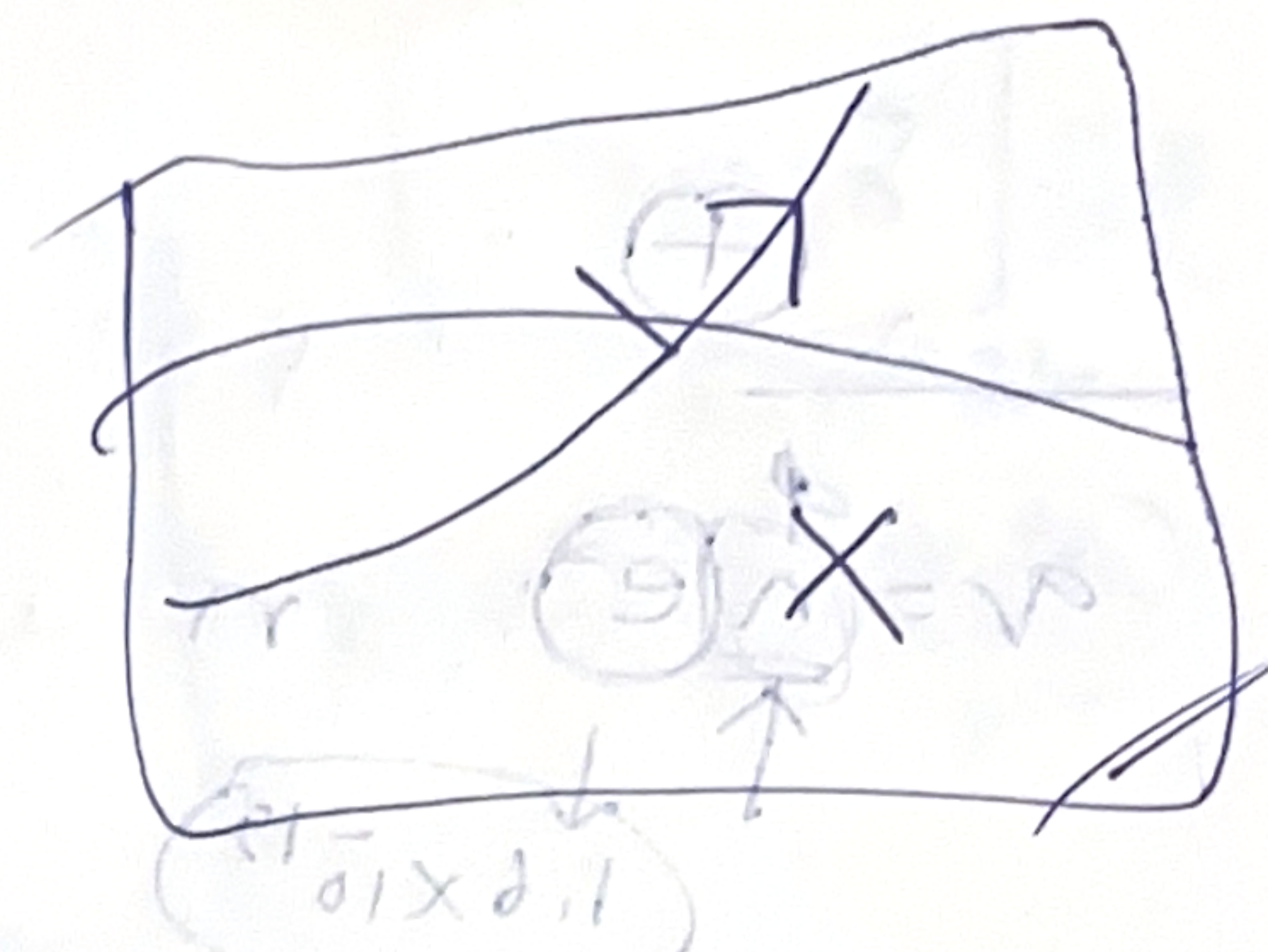
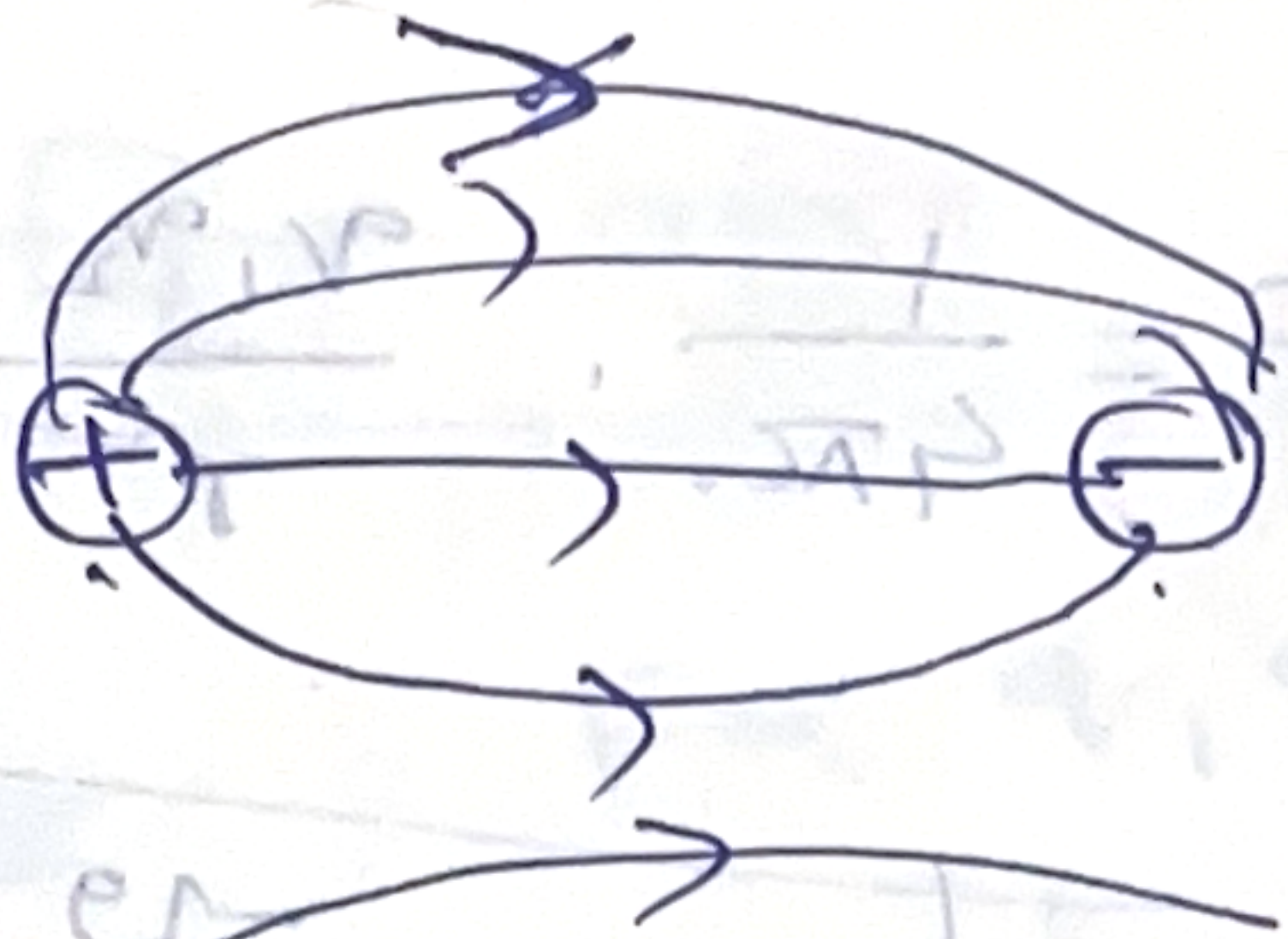


(+)

(+)

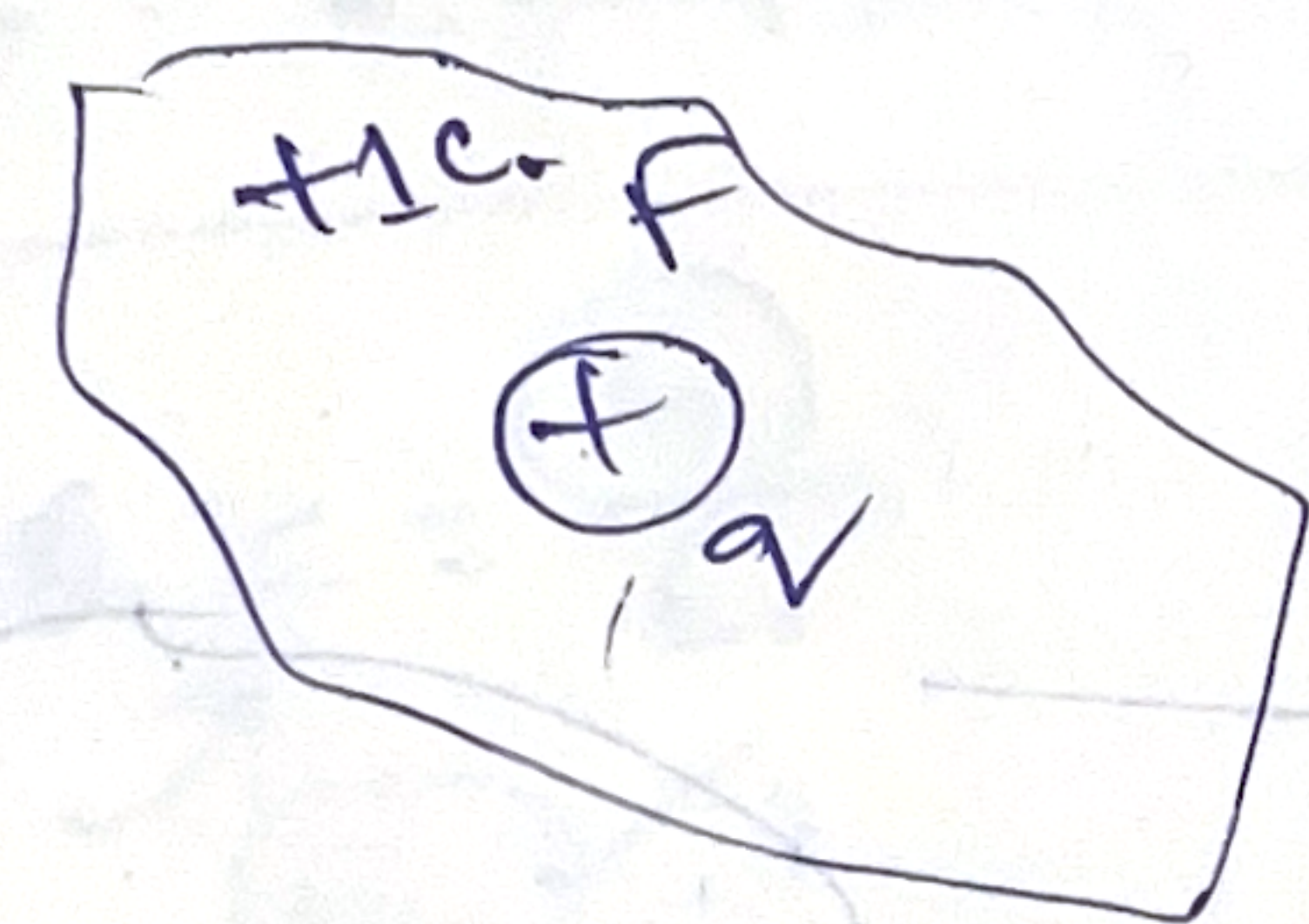
(+)

V/m

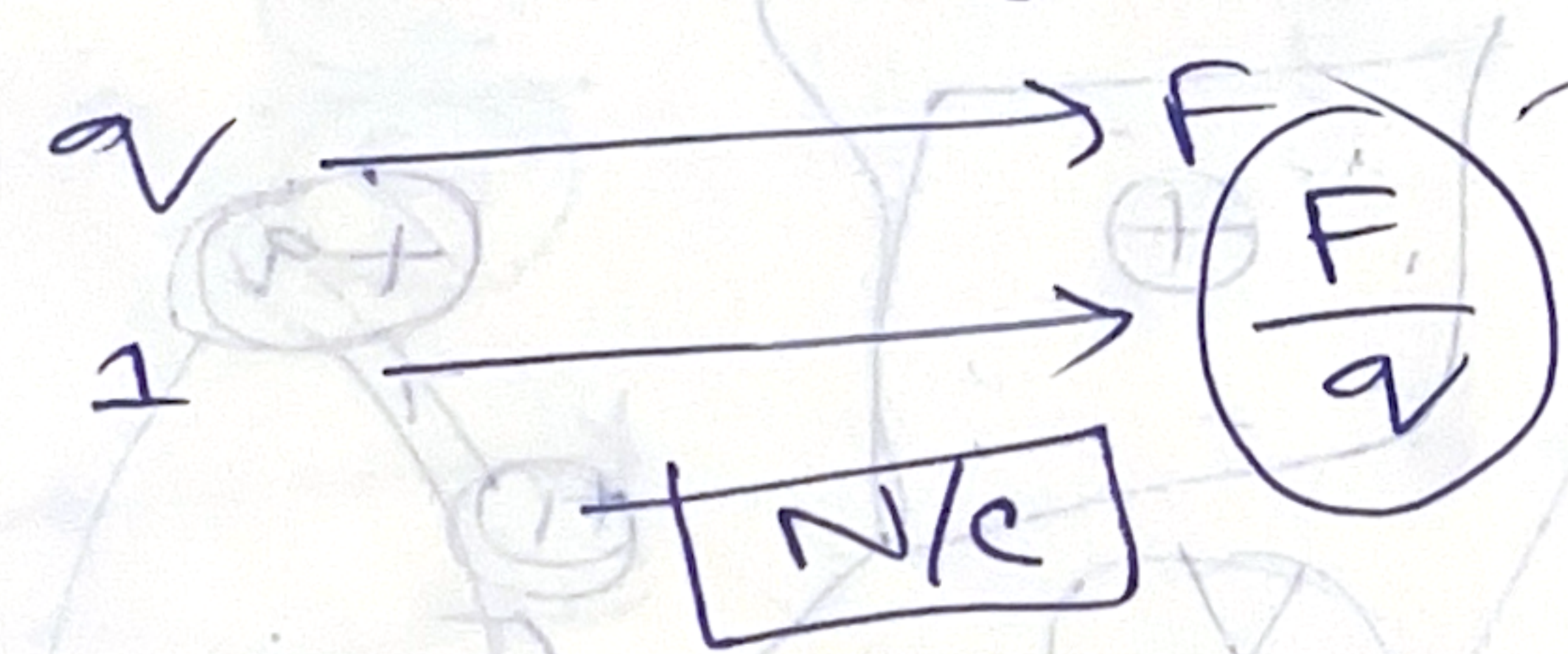


$\oplus \quad \ominus \quad \oplus$

$\oplus$ $\ominus$



$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$



$$E = (4\pi\epsilon_0) \cdot \frac{q}{r^2}$$

ଗଠିତ ଗୁଣିତ / ଗଠିତ ଗୁଣିତ

$$E = \frac{V}{d}$$

$$V/m$$



$$\vec{F} = q' \vec{E}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

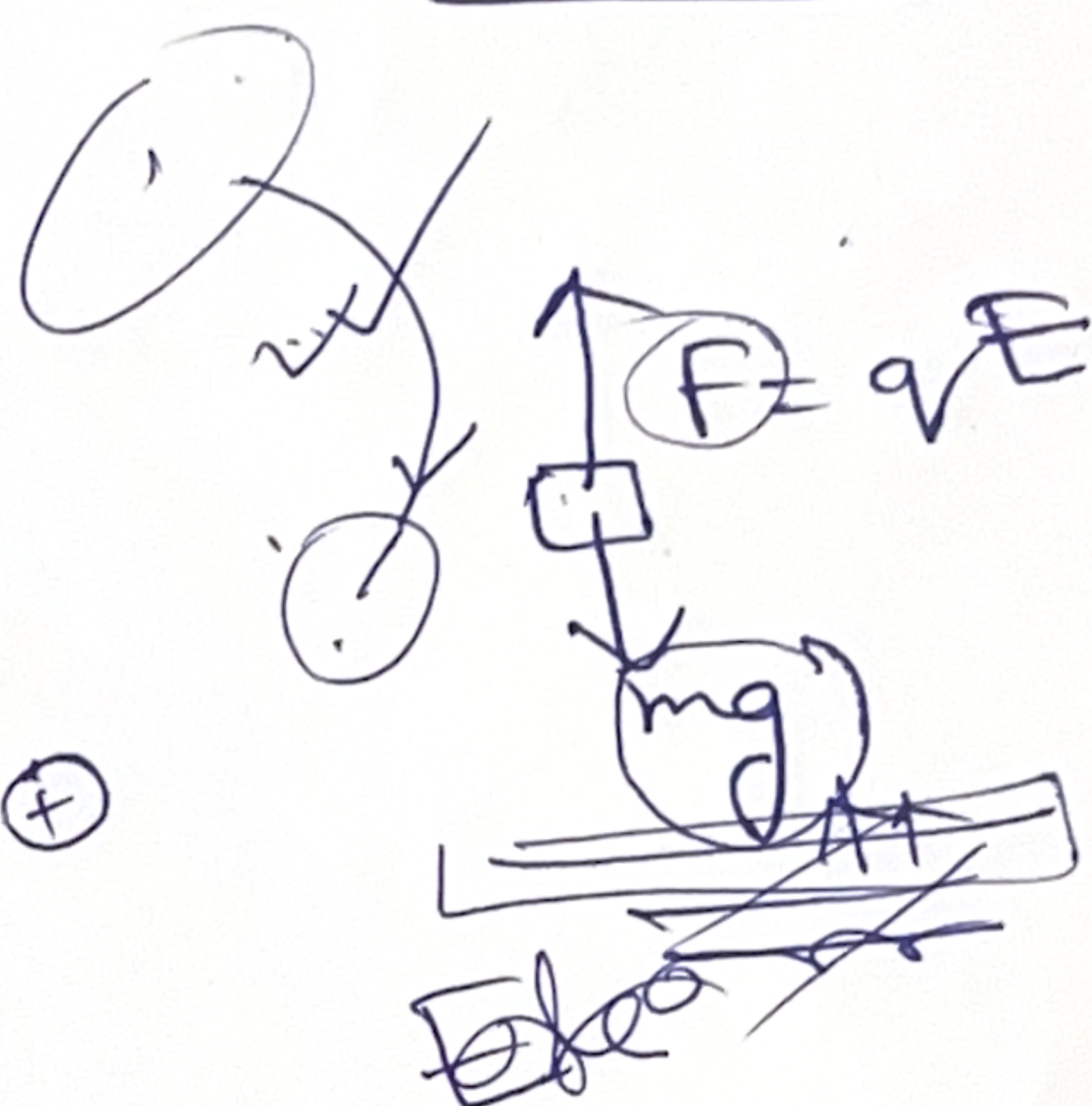
$$1 \longrightarrow \left(\frac{V}{A} \right) = \sigma$$

$$C m^{-2}$$

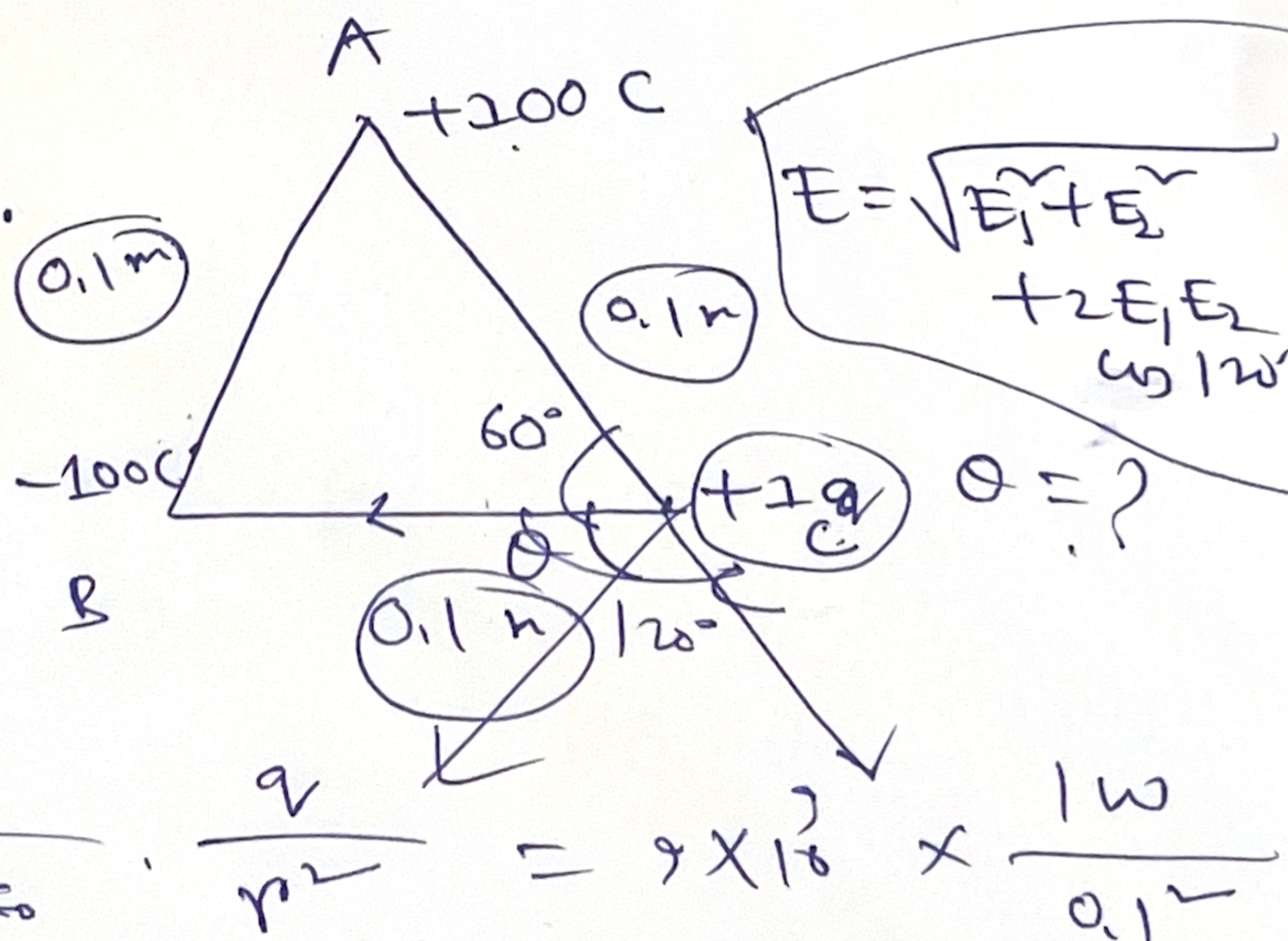
$$m = 5 \times 10^5 \text{ kg}$$

$$qE = mg$$

$q_v = ?$



9 नं०



$$E = \frac{2}{4\pi\epsilon_0} \cdot \frac{q}{r^2} = 9 \times 10^9 \times \frac{10}{0.1^2}$$

$= () ;$ CB zero

$$E = 9 \times 10^9 \times \frac{10}{0.12} = () ; \text{ A cm}$$